NI USRP-2954R FPGA/SDR Project Summary

OFDM Power Peak Position Modulation (PPPM) Receiver Study

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1 Project Overview

The purpose of this project was to investigate whether parts of an existing OFDM Power Peak Position Modulation (PPPM) communication experiment could be moved from a GNU Radio/Python implementation to the FPGA of an NI USRP-2954R.

The main motivation was to explore whether receiver-side signal processing could be performed directly on the USRP FPGA instead of relying entirely on the host computer. The expected advantages were more deterministic real-time processing, reduced host-side processing, and less need to transfer intermediate data between the USRP and the host.

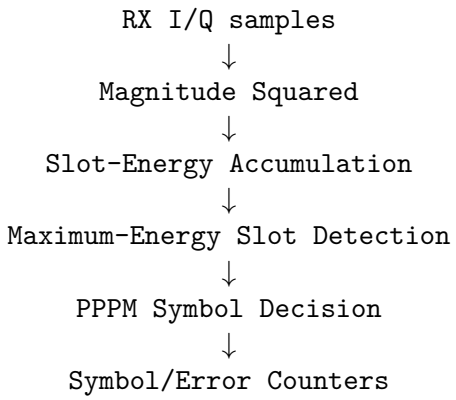
The project did not progress to a complete FPGA implementation on the physical USRP-2954R. The main reason was not a technical failure of the proposed design, but the licensing and software requirements associated with the NI/LabVIEW FPGA development environment. The additional cost was considered too high relative to the expected benefit of completing the prototype, and the FPGA implementation was therefore discontinued.

This document summarizes what was designed, what was not completed, and what could be reused if the work is continued in the future.

2 Proposed FPGA Architecture

The first FPGA version focused only on the core receiver energy-detection path. It assumed that the incoming payload samples were already correctly aligned.

The planned processing chain was:



The first version used fixed parameters:

- Eight PPPM slots per symbol.
- Sixty-four samples per slot.
- A total of 512 samples per PPPM symbol.
- A total of 1000 payload symbols per experiment.
- Integer/fixed-point FPGA arithmetic.

The architecture was split into four blocks.

2.1 Magnitude-Squared Block

The first block was intended to calculate the instantaneous power of each incoming complex I/Q sample. The design assumed signed I/Q samples and integer arithmetic suitable for FPGA implementation.

2.2 Slot-Energy Accumulator

The second block was intended to accumulate the sample power over 64 samples to generate one energy value for each PPPM slot.

2.3 Peak-Slot Detector

The third block was designed as a streaming maximum detector. It would process the eight slot-energy values and return the slot with the highest energy without needing to store the full energy vector in memory.

2.4 PPPM Decoder and Counters

The final block was intended to convert the detected slot into the corresponding PPPM symbol and update the experiment statistics. The main outputs were the number of processed symbols, the number of symbol errors, the last detected slot, the last decoded symbol, and a completion flag.

The symbol error rate itself was intended to be calculated on the host computer.

3 Host/FPGA Interaction

A simple register map was also prepared so that the host application could control the FPGA experiment and read the results.

Table 1: Main proposed host/FPGA registers.

Register	Direction	Purpose
`run_enable`	Host → FPGA	Start or stop receiver processing.
`reset_counters`	Host → FPGA	Reset internal state and counters.
`payload_symbols`	Host → FPGA	Number of symbols to process.
`symbol_count`	FPGA → Host	Number of processed symbols.
`error_count`	FPGA → Host	Number of symbol errors.
`last_slot`	FPGA → Host	Most recently detected peak slot.
`last_symbol`	FPGA → Host	Most recently decoded symbol.
`done`	FPGA → Host	Indicates that the run is complete.

The intention was for the host to configure and start the experiment while the FPGA handled the repetitive sample-level processing.

4 Expected Benefits

The FPGA approach was expected to provide several advantages compared with processing the complete receiver chain on the host.

- **Deterministic Processing:** Sample, slot, and symbol operations could be controlled directly by FPGA logic.
- **Reduced Host Workload:** Repetitive power calculation, accumulation, peak detection, and symbol counting could be removed from the CPU.
- **Reduced Intermediate Data Transfer:** Raw or intermediate sample information could be reduced to symbol decisions and statistics before reaching the host.
- **Streaming Operation:** The receiver blocks could operate continuously as a hardware pipeline.
- **Future Extension:** Synchronization, CFAR-style detection, and automated experiment control could later be added to the same FPGA framework.

These advantages were not measured experimentally in this project. No latency, throughput, CPU-load, or bandwidth comparison was completed, so they should be treated as expected architectural benefits rather than measured performance improvements.

5 Project Outcome

The main work completed was the FPGA design and planning stage.

The following were prepared:

- The first receiver architecture.
- Specifications for the four main FPGA blocks.
- Fixed-point and data-width planning.
- Host/FPGA register mapping.
- Block-level test ideas.
- A staged implementation and validation plan.

The following were **not** completed:

- A complete LabVIEW FPGA receiver.
- FPGA synthesis and compilation.
- A final bitfile.
- Physical USRP-2954R validation.
- End-to-end comparison against the original GNU Radio/Python receiver.

- Preamble and guard handling.
- Frame synchronization.
- CFAR-style or robust peak validation.
- TX-side FPGA frame generation.
- Measured FPGA performance improvements.

The project should therefore be described as an FPGA design and feasibility study rather than a completed FPGA implementation.

6 Why the Project Was Discontinued

The main limitation encountered was the NI FPGA development environment.

The USRP-2954R can be customized through LabVIEW FPGA, but continuing the project required access to the relevant NI software and licensing. For the scope of this experiment, the additional cost was considered too high compared with the expected research benefit of completing the prototype.

Because of this, the project was not taken through the full implementation, compilation, and hardware-validation stages.

This was primarily a practical and economic decision. The basic receiver operations themselves are conventional FPGA functions and no fundamental technical problem was identified in the proposed architecture.

If the required NI FPGA development environment is already available in the future, the existing design notes could be used as a starting point. Alternatively, the same processing architecture could be transferred to another FPGA/SDR platform with a more accessible development toolchain.

7 Limitations and Possible Continuation

The main technical limitation of the proposed first version is that it assumes the payload is already aligned. It does not detect the start of the frame or determine the correct slot timing.

A complete receiver would still require:

- Preamble detection.
- Guard handling.
- Frame and slot synchronization.
- Robust peak or CFAR-style validation.
- Comparison against the original GNU Radio/Python implementation.
- FPGA resource and timing analysis.
- Physical loopback or over-the-air testing.

If the project is resumed, the recommended first step is to recover the complete GNU Radio/Python reference model and validate the four proposed FPGA blocks against the same input data before adding synchronization or other advanced functions.

8 Conclusion

The project investigated the use of the NI USRP-2954R FPGA for accelerating parts of an OFDM-PPPM receiver originally implemented using GNU Radio and Python.

A simple streaming receiver architecture was designed around magnitude-squared calculation, slot-energy accumulation, peak-slot detection, PPPM decoding, and experiment counters. A host/FPGA register interface and a staged validation plan were also prepared.

The architecture suggests that these operations are suitable for FPGA implementation and could reduce host processing while providing more deterministic real-time execution. However, the project did not progress far enough to experimentally verify these benefits.

Development was discontinued mainly because the NI/LabVIEW FPGA licensing and toolchain requirements made the cost of completing the prototype difficult to justify for the expected research benefit.

The useful outcome of the project is therefore the documented receiver architecture and implementation plan. These materials can be reused if the NI environment becomes available in the future or if the receiver is migrated to another FPGA/SDR platform.

Project Files

The main retained handoff files are:

- Design specification: `v1_RX_Energy_Detector_Spec.md`.
- Magnitude-squared block: `v1_block_1_magnitude_squared.md`.
- Slot-accumulator block: `v1_block_2_slot_accumulator.md`.
- Slot-buffer/argmax block: `v1_block_3_slot_buffer_argmax.md`.
- PPPM-decoder/SER-counter block: `v1_block_4_pppm_decoder_ser_counter.md`.
- Top-level receiver design: `v1_top_level_rx_energy_detector.md`.
- Register map: `register_map.md`.
- NI USRP-2954R hardware documentation.

References

[1] National Instruments, *USRP-2954 Features*, May 2026.

[2] National Instruments, *USRP-2954 Specifications*, May 2026.

[3] National Instruments, *USRP-2950/2952/2953/2954/2955 Getting Started Guide*, May 2026.